

FRED Economic Data Modeling

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June 2026

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1 Introduction

The Federal Reserve Economic Data (FRED) database is a repository of economic data maintained by the Research Division of the Federal Reserve Bank of St. Louis. Established in 1991, FRED has become the global database for macroeconomic research and contains almost one million datasets from multiple regions and institutional sources. FRED streamlines data collection processes and improves the transparency of macroeconomic decisions.

With this data, I created a Vector Autoregression (VAR) model to understand the correlation and predictive power of CPI, S&P500, 10 Year Treasury, and Unemployment data on one another. Below are the definitions of each predictor used:

Variable 1: Unemployment Rate (UNRATE)

- **Official Series Name:** Unemployment Rate
- **Primary Sourcing Agency:** U.S. Bureau of Labor Statistics (Release: *Employment Situation*)
- **Definition:** The unemployment rate represents the number of unemployed as a percentage of the labor force. Labor force data are restricted to people 16 years of age and older, who currently reside in 1 of the 50 states or the District of Columbia, who do not reside in institutions (e.g., penal and mental facilities, homes for the aged), and who are not on active duty in the Armed Forces. This rate is also defined as the U-3 measure of labor underutilization. The series comes from the ‘Current Population Survey (Household Survey)’.
- **Citation:** U.S. Bureau of Labor Statistics, Unemployment Rate [UNRATE], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/UNRATE>.

Variable 2: Consumer Price Index (CPIAUCSL)

- **Official Series Name:** Consumer Price Index for All Urban Consumers: All Items in U.S. City Average
- **Primary Sourcing Agency:** U.S. Bureau of Labor Statistics (Release: *Consumer Price Index*)
- **Definition:** The Consumer Price Index for All Urban Consumers: All Items (CPIAUCSL) is a price index of a basket of goods and services paid by urban consumers. Percent changes in the price index measure the inflation rate between any two time periods. The most common inflation metric is the percent change from one year ago. It can also represent the buying habits of urban consumers. This particular index includes roughly 88 percent of the total population, accounting for wage earners, clerical workers, technical workers, self-employed, short-term workers, unemployed, retirees, and those not in the labor force.

The CPIs are based on prices for food, clothing, shelter, and fuels; transportation fares; service fees (e.g., water and sewer service); and sales taxes. Prices are collected monthly from about 4,000 housing units and approximately 26,000 retail establishments across 87 urban areas. To calculate the index, price changes are averaged with weights representing their importance in the spending of the particular group. The index measures price changes (as a percent change) from a predetermined reference date. In addition to the original unadjusted index distributed, the Bureau of Labor Statistics also releases a seasonally adjusted index. The unadjusted series reflects all factors that may influence a change in prices. However, it can be very useful to look at the seasonally adjusted CPI, which removes the effects of seasonal changes, such as weather, school year, production cycles, and holidays.

The CPI can be used to recognize periods of inflation and deflation. Significant increases in the CPI within a short time frame might indicate a period of inflation, and significant decreases in CPI within a short time frame might indicate a period of deflation. However, because the CPI includes volatile food and oil prices, it might not be a reliable measure of inflationary and deflationary periods. For a more accurate detection, the core CPI (CPILFESL) is often used. When using the

CPI, please note that it is not applicable to all consumers and should not be used to determine relative living costs. Additionally, the CPI is a statistical measure vulnerable to sampling error since it is based on a sample of prices and not the complete average.

- **Citation:** U.S. Bureau of Labor Statistics, Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL>.

Variable 3: Interest Rate Spread (T10YFFM)

- **Official Series Name:** 10-Year Treasury Constant Maturity Minus Federal Funds Rate
- **Primary Sourcing Agency:** Federal Reserve Bank of St. Louis (utilizing raw feeds from the Board of Governors of the Federal Reserve System and the U.S. Department of the Treasury)
- **Definition:** This series is calculated as the spread between the 10-Year Treasury Constant Maturity (BC_10YEARM) and the Effective Federal Funds Rate (<https://fred.stlouisfed.org/series/EFFRM>). Starting with the update on June 21, 2019, the Treasury bond data used in calculating interest rate spreads is obtained directly from the U.S. Treasury Department.
- **Citation:** Federal Reserve Bank of St. Louis, 10-Year Treasury Constant Maturity Minus Federal Funds Rate [T10YFFM], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/T10YFFM>.

Variable 4: Equity Market Index (S.P.500)

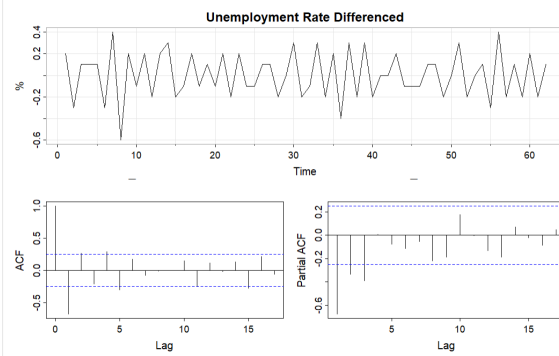
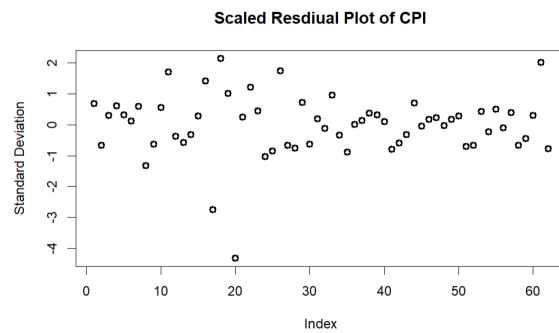
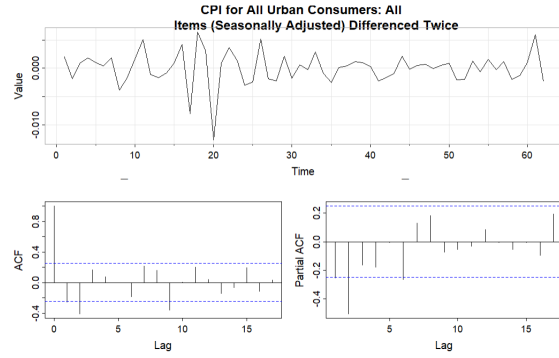
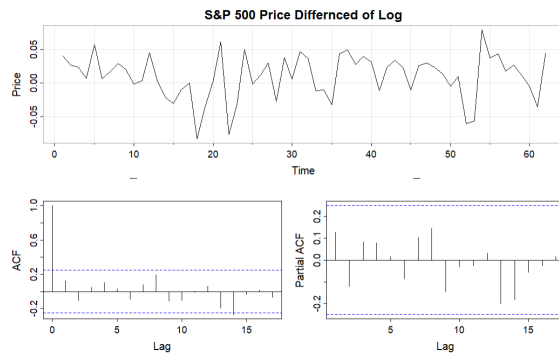
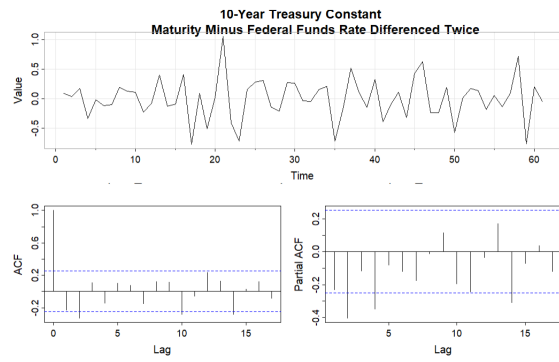
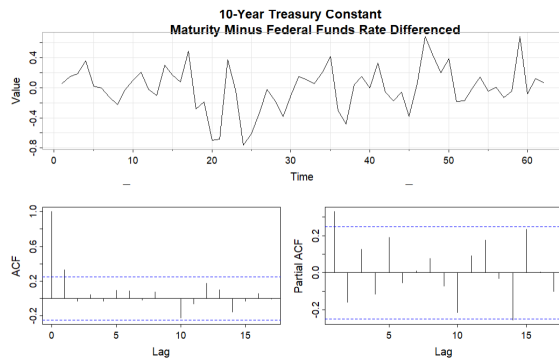
- **Official Series Name:** S&P 500
- **Primary Sourcing Agency:** S&P Dow Jones Indices LLC (Release: *Standard & Poor's*)
- **Definition:** The observations for the S&P 500 represent the daily index value at market close. The market typically closes at 4 PM ET, except for holidays when it sometimes closes early. The S&P 500 is regarded as a gauge of the large cap U.S. equities market. The index includes 500 leading companies in leading industries of the U.S. economy, which are publicly held on either the NYSE or NASDAQ, and covers 75% of U.S. equities. Since this is a price index and not a total return index, the S&P 500 index here does not contain dividends.
- **Citation:** S&P Dow Jones Indices LLC, S&P 500 [SP500], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/SP500>.

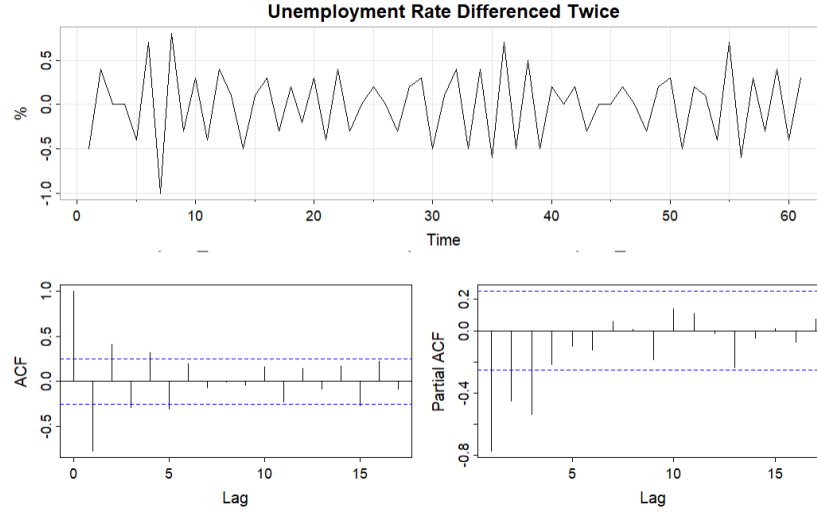
2 Data Preprocessing

The preprocessing starts with importing the raw dataset which is filtered into a target dataset that consisted of five variables: calendar date, 10-Year Treasury Constant Maturity Spread over the Federal Funds Rate, the Standard & Poor's 500 Index, the Consumer Price Index for All Urban Consumers, and Unemployment Rate. Manual calculation of CPI data was done for October 2025 due to a gap in data reporting caused by a government shutdown and Chicago Fed data was used for Unemployment Rate for the same month. These numbers were 323.8 and 4.4 respectively.

To achieve stationarity, I used McCracken and Ng FRED-MD database codes to transform the data to achieve stationarity. There were some caveats to their standard codes (see EDA). Transformations included first and second differences, natural log levels, and various log-differencing to isolate growth rates or accelerated changes. The data was cutoff at October 1, 2020 to evaluate relevant economic data post-COVID volatility.

3 Exploratory Data Analysis





The EDA plots show useful insight into the stationary properties, integration orders, underlying correlation structures for the variables. Stationarity is an integral part to proper time series models. For data to be stationary, it must have a constant variance and mean throughout. Meaning that there are no increases or decreases over time. For the transformations, the McCracken and Ng codes were used and then followed by analysis to determine if there needed to be another adjustment.

When looking at the 10-Year Treasury Spread, the differenced data appears to have a slight increase in mean over time from lag 20 to lag 60 while having flat valleys and sharp peaks in variance. This called for another differencing which appears to help with the keeping the mean more constant throughout and lowering the amount of variance spikes. The S&P500 log-differenced data shows a constant mean and some slight variance peaks, but nothing significant especially when looking at the ACF and PACF plots. The CPI's twice differenced data already came with a seasonal adjustment pre-transformation. This data has a constant mean with some sharp peaks around lag 20. I investigated these peaks to determine a potential adjustment to the transformation by fitting a linear regression to the CPI data. There were only two residual values greater than two standard deviations away which is fairly aligned with 68-95-99.7 rule. The Unemployment Rate differenced data showed to have a constant mean and variance over time. The second difference was to look to check on the variance as there are many sharps movements inside the differenced data. The second difference appeared to not change anything and the once differenced was used.

4 Model Selection

A Vector Autoregression (VAR) model is a multivariate time series framework that is used to capture linear inter-dependencies among multiple time series variables. A VAR model treats all variables symmetrically as endogenous system components. The variables consists of their own equations that express it as a linear function of its own historical lags, and a constant trend. A VAR(p) model tracking a vector of K edogenous varibales (Y_T) up to a lag order p is written as:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + \epsilon_t$$

Where c represents a vector of intercept constant, $A_1, \dots, A_p$ are $K \times K$ matrices of coefficients to be estimated, and ϵ_t is a vector of of white noise error terms.

Table 1: Comparative Estimation and Causality Summary of VAR(1) and VAR(2) Models

Equation / System Metric	Model 1: VAR(1)	Model 2: VAR(2)
<i>UNRATE Equation</i>		
Multiple R^2	0.4652	0.5336
Adjusted R^2	0.4270	0.4604
Residual Standard Error	0.1618	0.1556
<i>CPIAUCSL Equation</i>		
Multiple R^2	0.0724	0.4736
Adjusted R^2	0.0061	0.3911
Residual Standard Error	0.0029	0.0023
<i>T10YFFM Equation</i>		
Multiple R^2	0.2068	0.3298
Adjusted R^2	0.1502	0.2246
Residual Standard Error	0.2782	0.2674
<i>S.P.500 Equation</i>		
Multiple R^2	0.0411	0.1574
Adjusted R^2	-0.0274	0.0252
Residual Standard Error	0.0338	0.0332
<i>Global System Diagnostics & Causality</i>		
Lag Order (p)	1	2
Sample Size (T)	61	60
Log-Likelihood	419.551	439.334
Maximum Root Modulus	0.6795	0.7602
Breusch-Godfrey LM Test (p -value)	0.1470	0.1145
Granger Causality Test (p -value) [†]	0.9477	0.0346*
Instantaneous Causality Test (p -value) [†]	0.3707	0.9996

[†] Null Hypothesis (H_0): Financial markers (T10YFFM, S.P.500) do not cause Real indicators (UNRATE, CPIAUCSL).

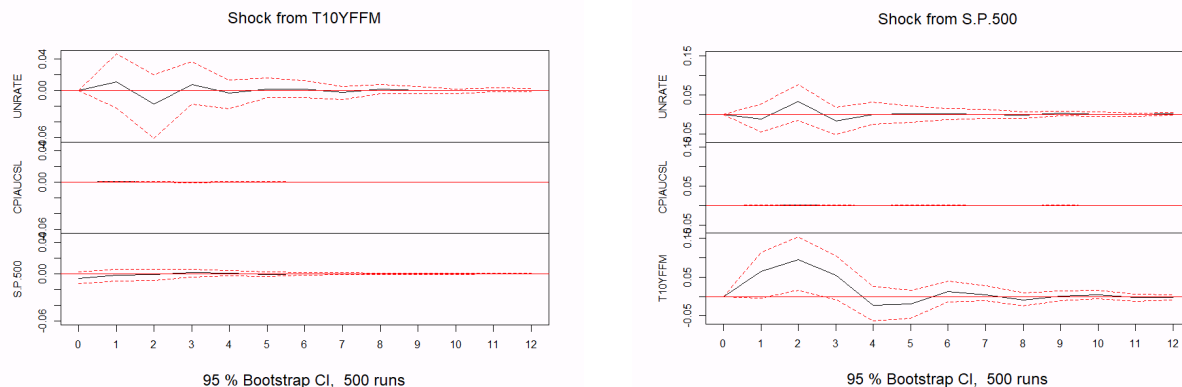
* Significant at the 5% level.

In evaluating the optimal lag configuration, a VAR selection process occurred to determine two possible candidate models. The *VARselect* function in R makes it simple to simultaneously use various model selection tests. The function output determined a VAR(2) and VAR(1) were plausible models to use according to AIC, BIC, Hannan-Quinn, and Final Prediction Error Criterion. AIC, HQ, and FPE favored VAR(2) due to the system's likelihood profile where increasing the lag length to two periods increases the Log-Likelihood from 419.551 in VAR(1) to 439.334 in VAR(2).

Restricting the system to a single lag showed severe underfitting. The Unemployment Rate achieves an adjusted R^2 of .4604 for VAR(2) and an adjusted R^2 of .427 for VAR(1). This jump in adjusted R^2 value happens due to there being a statistically significant second-order autoregressive lag. All the other variables see similar increases in adjusted R^2 values from the VAR(1) to VAR(2) model. Another compelling justification for selecting the higher-order lag structure is shown by the Granger causality tests. The Granger test evaluates whether historical information block from variables has predictive power over another block. Within the VAR(1) system, the test has non-significant p-value of 0.9477, failing to reject the null hypothesis. This implies that financial markets are completely decoupled from real economic outcomes. The VAR(2) specification has a statistically significant p-value of 0.03455. This allows us to reject the null hypothesis which establishes that financial variables do provide vital predictive leads for inflation and unemployment trajectories over a multi-month outlook. Tests for instantaneous causality show non-significant p-values across both the VAR(1) model ($p = 0.3707$) and the VAR(2) model ($p = 0.9996$). These results indicate a lack of significant zero-lag structural innovations flowing between the S&P500 market and economic sectors. Based on these diagnostics, the VAR(2) is selected as the optimal model. Removal of the second lag creates

an underfitted model that does not properly represent the data.

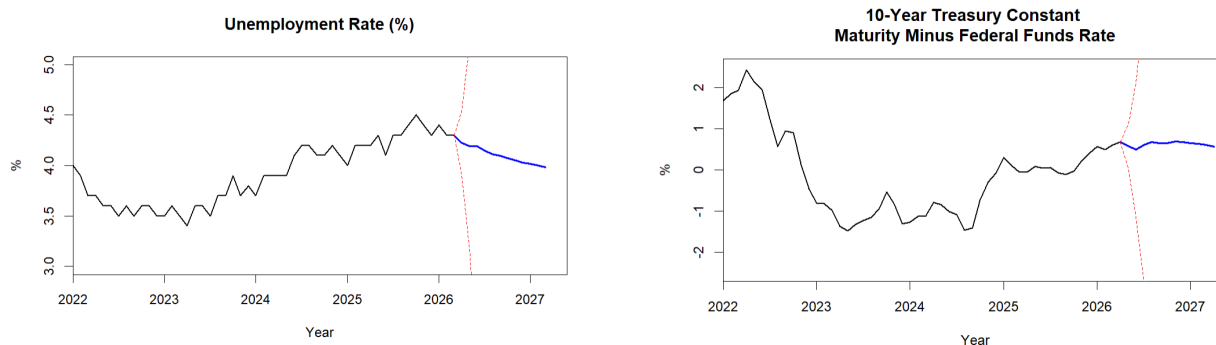
5 IRF

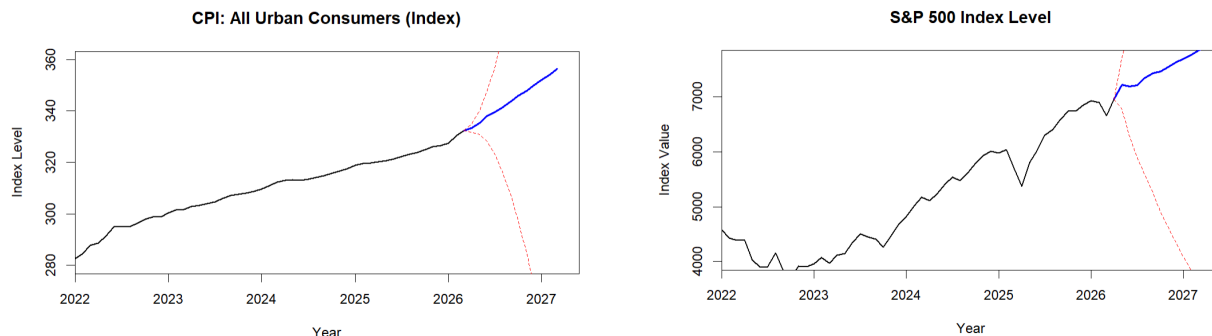


The Impulse Response Functions (IRFs) computed from the VAR(2) model shows the effects of an unexpected 1-standard-deviation shock from a variable to the rest of the system over a year. The solid black line represents the point estimates of the responses while the red lines show the lower and upper bounds of the 95% confidence interval. If the zero line is captured anywhere within the bands, the response is statistically indistinguishable from zero. This indicates that the shock fails to cause a net effect on that specific endogenous at that point in time.

The 10-Year Treasury Spread shows a weak and statistically insignificant effect on the broader economy. After a slightly positive increase in Unemployment Rate, there seems to be minimal movement throughout the rest while the S&P500 and CPI have no movement at all. The S&P500 shock shows to have more movement than the 10-Year Treasury Spread. The shock appears to cause opposite movement to the Unemployment Rate when compared to the 10-Year Treasury Spread, but it is still insignificant. The shock does have a significant increase at lag 2 for the 10-Year Treasury Spread. This indicates that Treasury yields tend to increase when there is a rapid increase in S&P500 price. This makes sense logically because it shows that when more cash is put into the stock market, treasury yields rise as a response to lower demand.

6 Forecasting





These plots show the one-year forecasts and associated confidence intervals of each variable. The blue line represents the expected increase/decrease in variable value while the dashed red line represents the confidence intervals (Note: Each variable was reversed-transformed back to its original raw values). The wide confidence intervals are due to the high volatility in macroeconomic markets. This shows the uncertainty of the macroeconomic world as well as difficulty to predict such data.

The Unemployment Rate plot shows a climb from 3.5% to 4.5% since 2023. Instead of the following this path the model projects a decrease in Unemployment Rate of about .3% throughout the next year. The CPI plot shows a very linear historical trajectory which climbs from about 280 and expects to march upward towards 360 by the middle of 2027. The 10-Year Treasury Spread shows a highly volatile history with values as high as 2% in 2022 that came with a crashing towards -1% in 2023. Even with this historically noisy series, the model projects a flat and stable path around .5% to .6%. The S&P500 plot shows steep rises and falls throughout with the overall increase being about 2500 points since 2022. The forecast projects a steady increase into the 7500 range throughout the next year. Collectively, these plots illustrate the nature of forecasting with the blue line showing a distinct trend, but the wide red bands serving as a reminder of how uncertain the macroeconomic world is.

7 Conclusion

This project investigated the dynamic inter-dependencies and predictive relationships between core macroeconomic indicators and financial markets by using a VAR model on post-COVID data. Preprocessing and exploratory data analysis was used to ensure the proper variables were being used, missing values were filled, and data was stationary. Model selection criteria favored a higher-order VAR(2) model over a VAR(1) alternative. Utilizing a second-order autoregressive lag yielded a substantial lift in R^2 across all variables by properly capturing delayed adjustment mechanisms. Granger causality testing confirmed that a single-lag model implies total structural decoupling, opposed to the VAR(2) model that shows a significant joint transmission channel. This confirms joint fluctuations in prices and yield curves contain predictive information in inflation and unemployment.

The Impulse Response Functions provided clear information regarding the effects of shocks on a specific variable. Shocks to the 10-Year Treasury Spread showed faint and statistically insignificant effects on the other variables. Alternatively, shocks on the S&P500 proved to have significant effects on the 10-Year Treasury Spread at lag two. This shows the rapid capital rotation from fixed-income into equities during periods of market optimism. Finally, the one-year forecasts gave insight into the possible trajectories of these markets. A cooling of labor markets, continuation of linear trend in consumer prices, a stable yield, and continued upward moment for equity prices were shown. These projections should be taken with caution as the large confidence intervals serve as a reminder of the immense variance inherent to macroeconomic forecasting.

8 Sources

References

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- [3] U.S. Bureau of Labor Statistics. (2026). *Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL]*, retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL>.
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- [6] McCracken, M. W., & Ng, S. (2016). *FRED-MD: A monthly database for macroeconomic research*. Journal of Business & Economic Statistics, 34(4), 574-589. Federal Reserve Bank of St. Louis Research Division. <https://www.stlouisfed.org/research/economists/mccracken/fred-databases>
- [7] Reuters. (2025, November 6). *US unemployment rate rounds up to 4.4% in October, Chicago Fed estimates*. Reuters Business News. <https://www.reuters.com/world/us/us-unemployment-rate-rounds-up-44-october-chicago-fed-estimates-2025-11-06/>

9 Tables

Table 2: Vector Autoregression (VAR(1)) Structural Estimation Results

Predictor	UNRATE	CPIAUCSL	T10YFFM	S.P.500
UNRATE.l1	-0.67384*** (0.09794)	-0.00009 (0.00178)	-0.04980 (0.16844)	-0.01839 (0.02049)
CPIAUCSL.l1	1.88424 (7.27882)	-0.26680* (0.13230)	25.04144 (12.51759)	1.26211 (1.52279)
T10YFFM.l1	0.02099 (0.07064)	0.00084 (0.00128)	0.33299** (0.12149)	0.00042 (0.01478)
S.P.500.l1	-0.27466 (0.64680)	0.00050 (0.01175)	2.26905* (1.11232)	0.16168 (0.13532)
const	0.00366 (0.02166)	0.00003 (0.00039)	-0.02554 (0.03725)	0.00808 (0.00453)
Multiple R^2	0.4652	0.0724	0.2068	0.0411
Adjusted R^2	0.4270	0.0061	0.1502	-0.0274
Residual Std. Error	0.1618	0.0029	0.2782	0.0338
F-Statistic	12.18***	1.092	3.650*	0.5998
Global System Diagnostics:				
Sample Size (T)	61			
Log-Likelihood	419.551			
Breusch-Godfrey LM	$\chi^2 = 212.59$, $df = 192$, p -value = 0.1470			
Characteristic Roots	0.6795, 0.3820, 0.2941, 0.1456			
Standard errors reported in parentheses below coefficient estimates.				
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

Table 3: Vector Autoregression (VAR(2)) Structural Estimation Results

Predictor	UNRATE	CPIAUCSL	T10YFFM	S.P.500
UNRATE.l1	-0.86944*** (0.12937)	0.00151 (0.00192)	-0.09075 (0.22225)	-0.05327 (0.02757)
CPIAUCSL.l1	1.59444 (7.32681)	-0.41987*** (0.10886)	17.69143 (12.58744)	1.98545 (1.56168)
T10YFFM.l1	0.03337 (0.07817)	0.00136 (0.00116)	0.40024** (0.13429)	-0.00213 (0.01666)
S.P.500.l1	-0.37015 (0.64618)	-0.00312 (0.00960)	1.97921 (1.11014)	0.16323 (0.13773)
UNRATE.l2	-0.32379* (0.12927)	0.00147 (0.00192)	-0.10216 (0.22208)	-0.04745 (0.02755)
CPIAUCSL.l2	-1.55019 (7.82448)	-0.55263*** (0.11626)	-24.18558 (13.44244)	2.93135 (1.66776)
T10YFFM.l2	-0.02955 (0.07267)	0.00105 (0.00108)	-0.10770 (0.12485)	-0.00469 (0.01549)
S.P.500.l2	0.70239 (0.66980)	0.02937** (0.00995)	1.82118 (1.15072)	-0.11828 (0.14277)
const	0.00113 (0.02166)	-0.00022 (0.00032)	-0.04207 (0.03721)	0.00925 (0.00462)
Multiple R^2	0.5336	0.4736	0.3298	0.1574
Adjusted R^2	0.4604	0.3911	0.2246	0.0252
Residual Std. Error	0.1556	0.0023	0.2674	0.0332
F-Statistic	7.292***	5.736***	3.137**	1.191
Global System Diagnostics:				
Sample Size (T)	60			
Log-Likelihood	439.334			
Breusch-Godfrey LM	$\chi^2 = 215.83$, $df = 192$, p -value = 0.1145			
Characteristic Roots	0.7602, 0.7602, 0.6028, 0.6028, 0.4229, 0.4229, 0.2285, 0.08599			

Standard errors reported in parentheses below coefficient estimates.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1